

CLT and Confidence Interval for μ

STAT 521

Applied Probability and Statistical Inference

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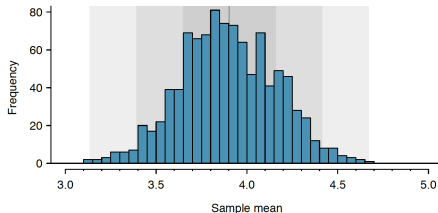
Variability in estimates

We have data of over 13,000 high school students taken in the Youth Risk Behavior Surveillance System (YRBSS) in 2013 and we know that the true average number of days a YRBSS student is physically active a week is $\mu = 3.9$ with standard deviation 2.56.

- Suppose that we take a random sample of 100 students and estimate that the average number of days a YRBSS student is physically active is $\bar{x} = 3.75$ days with a standard deviation of $s = 2.55$.
- Suppose we take another random sample of 100 individuals and take its mean: 3.22 days. Suppose we took another (3.67 days) and another (4.10 days), and so on.

Variability in estimates

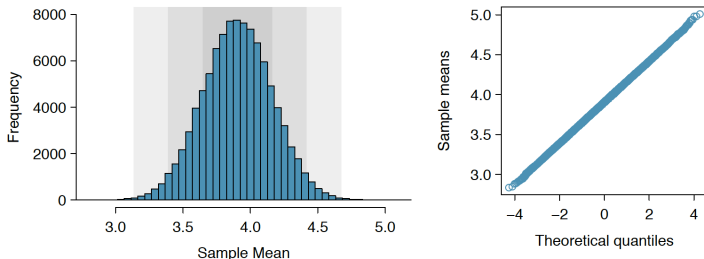
A histogram of 1000 sample means for number of days physically active per week for YRBSS data, where the samples are of size $n = 100$.



What can we say about the distribution of the sample means?

The sampling distribution for the mean

Histogram of the sample means for 100,000 different random samples of size $n=100$ for number of days physically active per week in YRBSS data . The right panel shows a normal probability plot of those sample means.



This empirically shows that the sample mean ($\bar{X}$) follows a Normal distribution.

Central Limit Theorem

Let $\bar{x}$ denote the sample mean computed from a random sample of n measurements from a population having a mean μ and finite standard deviation σ

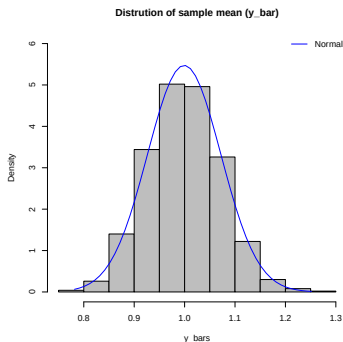
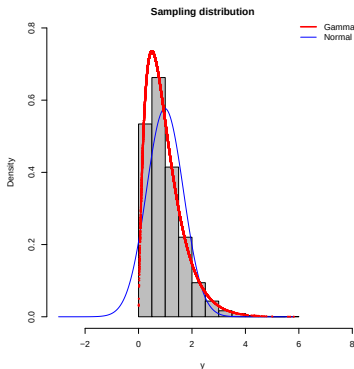
- 1 When n is large, the sampling distribution of $\bar{X}$ will be approximately normal with mean μ and standard deviation $\sigma/\sqrt{n}$: $\bar{X} \sim N(\mu, \sigma/\sqrt{n})$. The approximation is better as n increases.
- 2 When the population distribution is normal, the sampling distribution of $\bar{X}$ is exactly normal for any n .

Note that the CLT result requires a large sample size ($n > 30$) but does not require the data to be normally distributed. If the sample size is not large enough we need to verify normality to use it.

Example: Central Limit Theorem

What if the sampling distribution of the data is not Normal?

- Data from Gamma distribution, $Y_i \sim \text{Gamma}(2, 2)$, with $\mu = 1$ and $\sigma = 0.7$.
- Distribution of $\bar{Y}$ for $n = 100$ is then $\bar{Y} \sim N(1, 0.7/10)$.



Example: Stats final scores

Each year about 1500 students take the introductory statistics course at a large university. This year scores on the final exam are distributed with a median of 74 points, a mean of 70 points, and a standard deviation of 10 points. There are no students who scored above 100 (the maximum score attainable on the final) but a few students scored below 20 points.

- Is the distribution of scores on this final exam symmetric, right skewed, or left skewed?

Left skewed

- Would you expect most students to have scored above or below 70 points?

Above 70 points.

Example Cont.: Stats final scores

- Can we calculate the probability that a randomly chosen student scored above 75 using the normal distribution?

No.

- What is the probability that the average score for a random sample of 40 students is above 75?

In this case, $\mu = 70$ and $sd = 10/\sqrt{40} = 1.58 \rightarrow$

$$Z = \frac{75 - 70}{1.58} = 3.16$$

$$P(\bar{X} > 3.16) = 0$$

- How would cutting the sample size in half affect the standard deviation of the mean? *The sd would be bigger.*

Confidence intervals

- A plausible range of values for the population parameter is called a *confidence interval*.
- Using only a point estimate to estimate a parameter is like fishing in a murky lake with a spear, and using a confidence interval is like fishing with a net.



- If we report a point estimate, we probably will not hit the exact population parameter. If we report a range of plausible values – a confidence interval – we have a good shot at capturing the parameter.

Sampling Distribution of the Mean

Recall from the Central Limit Theorem that if we have a population with mean μ and standard deviation σ , and if either:

- the population is normal, or
- $n \geq 30$,

Then the sampling distribution of the mean is

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

We can standardize this distribution using

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

Then, Z follows the standard normal distribution $Z \sim N(0, 1)$.

Confidence Level and Confidence Interval Derivation

Let α be any number with $0 \leq \alpha \leq 1$. Given α , let $z_{\alpha/2}$ be the value such that the area under the standard normal curve between $-z_{\alpha/2}$ and $z_{\alpha/2}$ is $(1 - \alpha)$:

$$P(-z_{\alpha/2} \leq Z \leq z_{\alpha/2}) = 1 - \alpha$$

Replacing Z by its definition gives:

$$P\left(-z_{\alpha/2} \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq z_{\alpha/2}\right) = 1 - \alpha$$

Solving for μ :

$$P\left(\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$$

Confidence intervals for μ

- A confidence interval for the population mean can be computed as:

$$(\bar{x} - z_{\alpha/2}\sigma/\sqrt{n}, \bar{x} + z_{\alpha/2}\sigma/\sqrt{n})$$

where $z_{\alpha/2}$ is selected to correspond to the confidence level.

- The value $z_{\alpha/2}\sigma/\sqrt{n}$ is called the margin of error.
- The standard error for the sample mean is $SE_{\bar{X}} = \sigma/\sqrt{n}$

The confidence interval applies when:

- ① We have a simple random sample of size n ;
- ② The Central Limit Theorem holds (population normal or $n \geq 30$);
- ③ The population standard deviation σ is known.

Important Note

To construct a confidence interval, the first two conditions will always have to be satisfied.

However, the third condition is highly counterintuitive:

- We usually perform inference because the population parameters are unknown.
- Assuming that σ is known is often unrealistic.

In most cases, when the mean is unknown the standard deviation is also unknown $\rightarrow$ For large n ,

$$\bar{x} \pm z_{\alpha/2}s/\sqrt{n}???$$

A Short History of Student's t -distribution

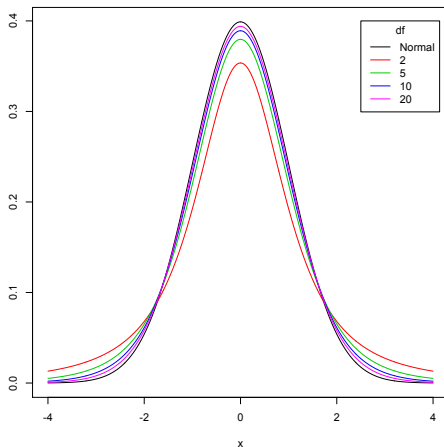
- The t -distribution was introduced in 1908 by William Sealy Gosset, an English statistician working at the Guinness Brewery in Dublin.
- Because Guinness forbade employees from publishing under their own names, Gosset wrote under the pseudonym “Student”.
- His paper, “*The Probable Error of a Mean*”, appeared in the journal *Biometrika*.
- Gosset developed the distribution to handle *small-sample inference* when the population standard deviation is unknown.
- The t -distribution is now one of the foundations of modern inferential statistics and hypothesis testing.

The t distribution

- The t distribution is a symmetric, bell-shaped distribution with heavier tails than the Normal distribution.
- The weight on the tails of the t distribution varies with the degrees of freedom (df).
- The degrees of freedom is generally the number of observations (or pieces of information) minus the number of parameters estimated.

The t distribution

As the degrees of freedom grow the t distribution converges to the Normal distribution.



Which critical value: z or t ?

We distinguish by whether the population standard deviation σ is known.

Case	Interval	Critical value
σ known	$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$	$z_{\alpha/2}$ (standard normal)
σ unknown	$\bar{x} \pm t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}$	$t_{\alpha/2, n-1}$ (Student t , df $n-1$)

Key point

If you replace σ with the sample standard deviation s , you must use the t distribution to account for the extra uncertainty:

$$\bar{x} \pm t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}.$$

What about large n ?

As $n \rightarrow \infty$, the t distribution approaches the standard normal:

$$t_{\alpha/2, n-1} \longrightarrow z_{\alpha/2}.$$

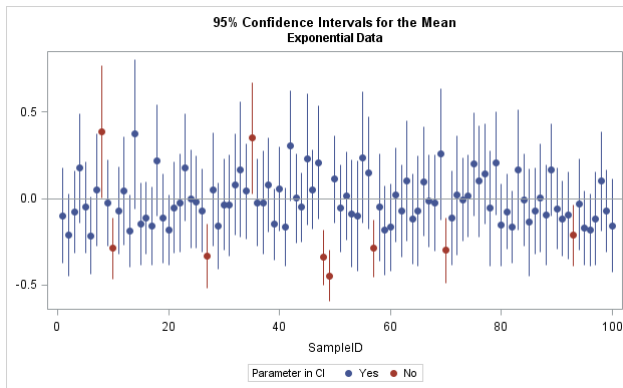
- For moderate n (e.g., $n \gtrsim 30$), $t_{\alpha/2, n-1} \approx z_{\alpha/2}$.
- For very large n (e.g., $n \gtrsim 100$), the difference is negligible.

Rule of thumb:

- Use t when σ is unknown (i.e., you plug in s).
- Using z is an excellent approximation for large n , but t is the exact choice.

Meaning of Confidence level

A level of confidence of 95% means that if we take repeated samples of size n and construct confidence intervals, we expect that approximately 5% of the intervals would fail to include the true population parameter.



Confidence level

- We are 95% confident that the interval $(\bar{x} - 1.96\sigma/\sqrt{n}, \bar{x} + 1.96\sigma/\sqrt{n})$ contains μ .

```
> qnorm(0.025)
[1] -1.959964
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- What's the Z-score for a 90% confidence interval?

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> qnorm(0.05)
[1] -1.644854
```

Estimating sleep

A survey asked students at GMU how many hours of sleep they get per night. A sample of 217 respondents yielded an average of 6.7 hours of sleep with a standard deviation of 2.03 hours. Assuming that this sample is random and representative of all students, we want to construct a plausible interval for the actual average amount of sleep students get per night.

There are certain assumptions and conditions that must be verified in order for the CLT to apply.

- *Random sampling condition:* We are assuming that this sample is random, and at least one of the following:
- *Approximate Normal Population:* If the population value is approximately normally distributed then the sampling distribution will also be approximately normal.
- *Large Sample:* If our sample is larger than 30 then the sampling distribution will be approximately normal

An approximate interval for sleep

The CLT then tells us that the sample mean of the hours of sleep of $n = 217$ GMU students will follow a normal distribution with mean μ and standard deviation $\sigma/\sqrt{n}$.

Since the sample size is large, the sampling distribution of the mean is approximately $N(\mu, s/\sqrt{n})$.

Based on this we can calculate an approximate 95% confidence interval for the average amount of sleep of GMU students as:

$$6.7 \pm \left(1.96 \times \frac{2.03}{\sqrt{217}} \right) = 6.7 \pm (1.96 \times 0.14) = (6.42, 6.98)$$

What does this mean?

We are 95% confident that GMU students sleep between 6.42 to 6.98 hours a night on average (the true population mean, μ is within this range).

Changing the confidence level

Confidence interval, a general formula

$$\text{point estimate} \pm z^* \times SE$$

- In order to change the confidence level all we need to do is adjust z^* in the above formula.
- Commonly used confidence levels in practice are 90%, 95%, 98%, and 99%.

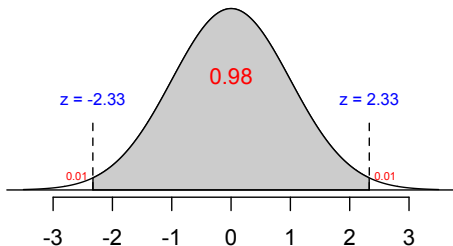
Confidence interval for μ for large n

$$\bar{x} \pm z_{\alpha/2} \times s/\sqrt{n}$$

A 98% confidence interval

Calculate a *98% confidence* interval for the average sleep GMU students get per night.

$$\bar{x} = 6.7, s = 2.03, n = 217$$



$$\begin{aligned} CI &= \bar{x} \pm z_{\alpha/2} \times s / \sqrt{n} \\ &= 6.7 \pm \left(2.33 \times \frac{2.03}{\sqrt{217}} \right) \\ &= (6.7 - 0.32, 6.7 + 0.32) \\ &= (6.38, 7.02) \end{aligned}$$

Which of the following is correct?

- Ⓐ 98% of GMU students sleep between 6.38 and 7.02 hours per night, on average.
- Ⓑ We are 98% confident that GMU students on average sleep 6.38 to 7.02 hours per night.
- Ⓒ *We are 98% confident that GMU students on average sleep 6.38 to 7.02 hours per night.*
- Ⓓ 98% of the time GMU students sleep 6.38 hours to 7.02 hours per night.
- Ⓔ We are 98% confident that the average sleep the 217 students in this sample get is between 6.38 and 7.02 hours per night.
- Ⓕ The standard error is 0.32 hours.

Testing claims based on a confidence interval

The 95% confidence interval for the average hours of sleep of GMU students was (6.43, 6.97). Does this provide convincing evidence that GMU students do not get the recommended 8 hours of sleep?

- Using a confidence interval for hypothesis testing might be insufficient in some cases since it gives a yes/no (reject/don't reject) answer, as opposed to quantifying our decision with a probability.
- Formal hypothesis testing allows us to report a probability along with our decision.